SHAYAN AMOUZADEH

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RESEARCH INTERESTS

- **Artificial Intelligence & Machine Learning:** Large Language Models (LLMs), Natural Language Processing (NLP), Multimodal Speech-to-Text Pipelines, and Information Retrieval.
- **Software Engineering & Computer Systems:** Automated Testing & Formal Verification, High-Performance & Deterministic Computing, Computer Graphics, and Secure Web Architectures.

EDUCATION

B.Sc. in Computer Engineering (Software Engineering Specialization) **Sep 2022 – Sep 2026**

Semnan University, Faculty of Electrical and Computer Engineering — Semnan, Iran

- **Academic GPA: Selected Core CS/CE Average:** 18.03 / 20.00 (21 Courses)
- **Final Year GPA:** 17.03 / 20.00
- **Last 2 Years GPA:** 16.93 / 20.00
- **Cumulative GPA:** 15.77 / 20.00
- **B.Sc. Thesis:** *Design & Implementation of an Intelligent Real Estate Management System*
  (**Grade: 19.50 / 20.00** • Advisor: Dr. Morteza Dorrigiv)
- **Selected Core Coursework (Grade / 20.00):** Information Technology (20.00), Programming Languages Design (19.50), Computer Networks Lab (19.50), Software Engineering (19.25), Operating Systems Lab (19.25), Web Programming (18.50), Linear Algebra (18.00), Data Structures & Algorithms (17.50).

UNDERGRADUATE THESIS & APPLIED AI RESEARCH

Developer & System Architect — Intelligent Real Estate CRM Platform **2025 – 2026**

(Live: https://amlak-e-man.ir)

Advisor: Dr. Morteza Dorrigiv, Semnan University • https://github.com/Shayan-Amz/Real-Estate-File-Manager-Project

- **Multi-Tenant Architecture & RBAC:** Architected and deployed a production real estate CRM (PHP/PDO, MySQL 7-table schema, 17 REST action handlers, and zero-dependency vanilla-JS PWA) featuring version-gated runtime schema migrations, composite indexing, a 15 s change-token synchronization protocol, and 3-role RBAC (*Manager* / *Consultant* / *Guest*) across 3 commercial tiers.
- **Human-in-the-Loop Voice & LLM Extraction Pipeline:** Chained browser 16 kHz mono PCM-16 WAV encoding with provider-switchable Whisper-large-v3 STT (Hugging Face, Groq, OpenAI) and an OpenRouter LLM extraction agent governed by a 16-rule Persian JSON contract, prompt-injection guardrails, a 26-field server allow-list, and client-side spoken-numeral/currency normalization into a human-reviewed modal.
- **Demand-Property Matching & Geospatial Integration:** Implemented a client-side matching engine ($O(P)$ scan per demand query across P active listings) using normalized bidirectional city matching, ±15% area tolerance, and 3-mode budget constraints ($\leq 1.10\times$), integrated with the lazy-loaded **Neshan (Iranian location service) Leaflet SDK** for interactive geocoding and deep-link routing.
- **Security Hardening & Automated Evaluation:** Designed 30-hour HMAC-SHA256 tenant-bound tokens verified in constant time, per-request consultant revocation, SHA-256 masked tenant IDs in guest responses, a 150 req/min sliding-window rate limiter, dual-stage image watermarking/sanitization, and 47 automated tests alongside an offline Persian WER/F1 evaluation console secured by HS256 JSON Web Tokens (JWT, RFC 7519).

SELECTED RESEARCH & OPEN-SOURCE PROJECTS

IEEE 754 Single-Precision Floating-Point Unit https://github.com/Shayan-Amz/Floating-Point-ALU

C++17 Header-Only (constexpr), CMake, 4-Way Cross-Platform CI — Hardware Arithmetic Verification

- Implemented a bit-exact, compile-time (constexpr) C++17 software model of an IEEE 754 binary32 arithmetic datapath (add, sub, mul, div: unpack → sticky-bit shift alignment / 48-bit product / 27-bit integer long division → normalization → Guard-Round-Sticky rounding → pack) supporting all 4 IEEE 754 rounding modes, subnormals, signed zeros, ±∞, quiet/signaling NaN propagation, and bitwise trace inspection.
- Verified correctness via differential testing against the host FPU under fesetround(): executed **32,026,992 operand-pair checks** (41 × 41 edge grid + 8 randomized stress distributions) with **0 mismatches**, enforced on a 4-way GCC/Clang/AppleClang/MSVC CI matrix compiled with -frounding-math and /fp:strict.
- Quantified defects in an initial course adder using a 1.4M-sample accuracy harness (58.2% bit-exact, 40.8% 1-ULP error, 1.1% > 1-ULP error) and achieved **100% bit-exact compliance** by adding sticky-bit alignment, round-to-nearest-even, and gradual-underflow handling.

Deterministic Graphics Pipeline & CPU Software Rasterizer https://github.com/Shayan-Amz/OpenGL-Dream-Village

C++17, C99, OpenGL 1.1, FreeGLUT, CMake, 3-OS GitHub Actions CI — Computer Graphics & Determinism

- Refactored an immediate-mode OpenGL 1.1 animated scene into a decoupled AnimationState and pure render() architecture compiled against both system OpenGL and custom headers for cross-platform FreeGLUT execution and headless GPU-less regression testing.
- Authored softgl, a C99 deterministic software rasterizer implementing 49 GL/GLUT entry points — Juan Pineda's (SIGGRAPH '88) parallel half-space edge functions in double precision, pixel-center sampling, top-left fill rule, polygon fan-triangulation, DDA line rasterization, and Gouraud shading — verified by 28 unit self-test assertions.
- Enforced pixel-exact CI regression (–max-diff-ppm 0) across 4 dynamic scenes (up to 7,017 triangles and 1.26M fragments/frame at 1300 × 650 resolution) on GCC, AppleClang, and MSVC, resolving an ARM64 FMA-contraction discrepancy via -ffp-contract=off and eliminating 605 per-frame legacy GL errors.

Bare-Metal ARMv7-A VGA Framebuffer Renderer https://github.com/Shayan-Amz/ARM-Assembly-Flag-Renderer

ARMv7-A GNU/Zig Assembly, Python Unicorn CPU Emulator, CI — Low-Level Systems & Emulation

- Wrote a bare-metal ARMv7-A (Cortex-A9 / Intel DE1-SoC) graphical renderer filling the memory-mapped 320 × 240 RGB565 VGA framebuffer at 0xC8000000 via an AAPCS-compliant leaf routine using caller-saved registers (r0–r3) with zero stack spills, packing two 16-bit pixels per 32-bit word store across the 1,024-byte row stride.
- Built a Python Unicorn-engine emulation harness executed in CI across-GNU binutils and Zig/Clang cross-assemblers to instrument every instruction and memory write, verifying exact stripe geometry (80/80/80 rows), clean halt detection, and zero out-of-bounds stores.
- Reduced executed instructions from **353,286 to 116,201 (**−67% **/** 3.04× **reduction)** and eliminated **9,216 out-of-bounds memory writes** and **46,080 redundant padding writes** compared to the initial baseline implementation.

End-to-End Box-Office Scraping & Content Recommender Pipeline https://github.com/Shayan-Amz/Movie-ML-Pipeline

Python 3.10+, Pandas, BeautifulSoup4, Streamlit, Pytest & Ruff CI — Data Engineering & Content Filtering

- Engineered a pip-installable src-layout data pipeline scraping 12 years (2013–2024 / 1392–1403 SH) of Iranian box-office archives into a clean **686-film × 52-column dataset** with outlier filtering, Persian numeral normalization, and **43 multi-label one-hot genre features** (102 unique genre signatures).
- Designed a deterministic content-based recommendation engine matching exact 43-bit genre signatures to deliver dual Top-K rankings by critic rating and box-office revenue inside a cached Streamlit UI, backed by **14 pytest unit tests** and ruff CI checks across Python 3.10 and 3.12.

Exact-Optimal Fractional Knapsack & DP Benchmark Suite https://github.com/Shayan-Amz/Fractional-Knapsack-Greedy

C++17 Header-Only, Python Benchmarking, 4-Way Cross-Platform CI — Algorithmic Optimization

- Implemented an $O(n \log n)$ header-only C++17 greedy fractional knapsack solver using a division-free cross-product value-density comparator ($v_a w_b > v_b w_a$) with std::stable_sort for deterministic tie-breaking, paired with an $O(nW)$ 1D rolling-array 0/1 dynamic programming reference solver, LP-relaxation upper bound for branch-and-bound, and a formal exchange-argument optimality proof.
- Validated optimality across **328,582 automated checks (0 failures)** — comparing against CLRS reference instances, 0/1 DP bounds, and exhaustive subset-and-fractional-item enumeration ($O(n \cdot 2^n)$ brute-force oracle) across 20,000 randomized instances — on a 4-way GCC/Clang/AppleClang/MSVC CI matrix; benchmarked the $O(n \log n)$ greedy solver at **1.86 ms vs. 26.38 s** for the $O(nW)$ 0/1 DP reference at n = 20,000 and characterized integrality gap decay from 30.0% (n = 5) to 0.05% (n = 200).

Full-Stack Stochastic World Cup Simulator & Analyzer https://github.com/Shayan-Amz/World-Cup-Simulator-And-Analyzer

JavaScript (ES2017), PHP 8.3 REST API, MySQL 8 / SQLite, Docker Compose — Stochastic Simulation & Full-Stack

- Developed a containerized 16-team tournament simulator (vanilla-JS SPA + PHP 8 REST API + MySQL/SQLite) over a **178-national-team rated dataset** (ratings 46–94 across 6 confederations), orchestrating a 24-match round-robin group stage and seeded 8-team knockout bracket (31 matches total) with points → goal difference → goals for tie-breaking.
- Designed a rating-driven stochastic match engine featuring a 25% draw prior, a 7-tier win-probability step function over rating gaps (Δ), categorical scoreline sampling, and asymmetric goal perturbations for $|\Delta| > 15$, characterized via a **100,000-match Monte-Carlo simulation study** and secured by bcrypt session authentication and native PDO prepared statements.

Professional & Industry Experience

Computer Engineer & IT Specialist **Mar 2025 – Present (Full-Time)**

Khakab Sazeh Ghomes Co. — Semnan, Iran

• **Enterprise IT & Network Administration:** Manage internal network infrastructure, hardware/software provisioning, and daily system administration across company operations.

• **Database & Project Automation:** Design and maintain relational databases and automated financial/engineering billing models using Microsoft Access, Excel, and Microsoft Project (MSP) to streamline project scheduling and resource tracking.

Game Developer & Software Engineering Intern **Jul 2025 – Dec 2025**

Sargarmi Sazan Aseman Omid (Game Development Studio) — Semnan, Iran

• **Data-Oriented Architecture (ECS):** Architected core gameplay mechanics in Unity (C#) utilizing the Entitas Entity Component System (ECS) framework on proprietary studio projects, strictly decoupling immutable component state from reactive execution systems.

• **UI & Performance Engineering:** Improved runtime frame-time consistency, modularity, and maintainability over conventional OOP inheritance hierarchies; additionally built open-source Unity 6 uGUI/URP 2D interfaces with LINQ-driven reactive search and sorting (https://github.com/Shayan-Amz/Unity-School-Directory-UI).

Technical Skills

• **Programming Languages:** C++ (C++17, STL, constexpr), Python 3, ARMv7-A Assembly, C#, PHP 8, JavaScript (ES2017+), SQL, C99, HTML5/CSS3.

• **Systems & Verification:** IEEE 754 Arithmetic Modelling, Differential FPU Testing (fesetround), Deterministic Software Rasterization, Unicorn CPU Emulation, Bare-Metal Memory-Mapped I/O (DE1-SoC VGA).

• **AI, NLP & Data Pipelines:** LLM Structured Extraction (OpenRouter, Prompt-Injection Guardrails), Whisper STT Audio Processing, Persian NLP Normalization, Pandas, BeautifulSoup4, Streamlit, WER/F1 Evaluation.

• **Web, Databases & Security:** RESTful APIs, Zero-Dependency PWA/SPA, Neshan Leaflet SDK, MySQL / MariaDB / SQLite (PDO, Schema Migrations), Stateless HMAC-SHA256 & HS256 JWT Auth (RFC 7519), Rate Limiting.

• **Graphics, Game Engines & DevOps:** OpenGL / FreeGLUT, Unity 6 (uGUI, URP 2D, LINQ), Entitas ECS, GitHub Actions (Multi-OS Matrix CI), CMake, GCC / Clang / MSVC, Docker, Pytest, Ruff, Puppeteer.

Languages

• **Persian:** Native Proficiency

• **English:** Advanced Working Proficiency (IELTS Academic in preparation).